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### SonarQube

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=> Code Quality Checking software or Code Review Software.

=> Using sonarqube we can perform code review to identify developers mistakes in the code.

=> SonarQube s/w developed by using Java language.

=> SonarQube supporting 30+ Programming languages code review.

Ex: C, C++, Python, Java, Dot net, SQL, Ruby, java script...

Note: We can use sonar trail version for practice.

=> In company we will use sonar enterprise version (commerical)

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### Sonar Issues

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=> sonarqube server will identify below issues in the project.

- 1) Bugs (danger, it may stop code execution)
- 2) Vulnerabilities (security hotspots)
- 3) Code Smells (not danger, but weak design)
- 4) Duplicate Code Blocks
- 5) Code Coverage (how many lines of code is tested)

Note : DevOps team is responsible to perform Code review and send code review report to Development team.

Note: Development team is responsible to fix the issues identified by sonar server.

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### Sonar Quality profiles

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=> Quality Profile means set of rules to perform code review.

=> In SonarQube for every language one quality profile available

java project ---> java quality profile ---> java language rules are configured

python project ---> python quality profile ---> python language rules are configured

dot net project ---> dotnet quality profile ---> dot net rules are configured

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### Sonar Quality Gate

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=> Quality Gate represents overall project code quality is Passed or Failed.

Note: If project quality gate is failed then we shouldn't deploy that code in server.

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## SonarQube Setup

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Setup Doc : <https://github.com/ashokitschool/DevOps-Documents/blob/main/06-Sonar-Setup-Docker.md>

Note: SonarServer runs on port number 9000. Enable it in Ec2 vm security group inbound rules.

default username : admin

default password : admin

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## SonarServer Integration in CICD Pipeline

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=> To integrate sonar server in jenkins pipeline we need sonar server token.

SonarServer login => Goto profile => My Account => Security => Generate Token.

Token : sqa\_f5794766e3aec9f61e85c37d6521e52797e03b0c

Note: In jenkins pipeline after maven build process completed we will perform Code Review using SonarQube server.